Polish-Japanese Academy of Information Technology

Bachelor's engineering thesis

Short-Term Forecasting of Martian Surface Weather Using Machine Learning

PRO3, Task 02: Chapter 2, Literature review

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2 Literature review

2.1 Scope and method of the review

This chapter reviews the literature on which the forecasting problem of Chapter 1 builds, organized by theme rather than chronologically. Sections 2.2 and 2.3 cover the observations and the physical models of Martian weather, Sections 2.4 and 2.5 data-driven forecasting on Mars and on Earth, and Sections 2.6 and 2.7 forecasting methods and the transfer of models between sites. Section 2.8 compares the approaches, identifies the research gap and positions the thesis.

Sources were identified by keyword searches in NASA ADS, Google Scholar, arXiv, IEEE Xplore and the ACM Digital Library, by following the references of the studies found and by a repeated search of the OpenAlex, Crossref and arXiv indexes in October 2026. The queries combined domain terms, such as "Mars surface meteorology", "REMS", "MEDA", "InSight" and "Martian boundary layer", with method terms, such as "time series forecasting", "gradient boosting", "temporal convolutional network", "machine learning weather prediction" and "transfer learning". Peer-reviewed articles, conference papers and books were included. Conference abstracts and preprints were included only where the work they report is not available in peer-reviewed form, which applies to much of the recent work on Mars, and they are identified as such in the text. Blogs, tutorials and studies that do not identify their data were excluded. Observations of Mars are reviewed from the Viking landers of the 1970s onwards and data-driven forecasting mainly from 2016 to 2026; older methodological works are retained where they define current practice, such as skill scores and out-of-sample testing.

2.2 Near-surface weather observed by landed stations

The two Viking landers made the first long meteorological records on the Martian surface. In the northern summer of their nominal missions, the diurnal patterns of pressure, temperature and wind were strongly similar from one sol to the next[1]. A review of all in situ records from Viking to Curiosity, spanning more than twenty Martian years, found the same seasonal pressure cycle at every site, driven by the exchange of about 30 % of the atmosphere with the polar caps. Global dust storms raised the diurnal pressure amplitude and reduced the diurnal amplitude of air temperature, and of all variables, wind depended most on local and regional topography[2].

Since 2012, REMS has sampled Gale Crater for five minutes every hour at 1 Hz, with additional full-hour blocks[3]. Over the first 100 sols, the diurnal pressure range averaged 85 Pa

[1] S. L. Hess et al. "Meteorological results from the surface of Mars: Viking 1 and 2". In: *Journal of Geophysical Research* 82.28 (1977), pp. 4559–4574.

[2] G. M. Martínez et al. "The Modern Near-Surface Martian Climate: A Review of In-situ Meteorological Data from Viking to Curiosity". In: *Space Science Reviews* 212.1–2 (2017), pp. 295–338.

[3] *Ibid.*

at a mean pressure of 775 Pa, more than at any earlier landing site[4]. Over a full Mars year, the diurnal tide had amplitudes of 26–63 Pa and the semidiurnal tide 5–20 Pa, both closely correlated with dust opacity[5]. Part of this range is local: global models with grid spacings of 120–500 km simulate only about half of it, and models with spacings below 5 km reproduce the rest as a hydrostatic adjustment of the air over the topography of the crater region[6]. Wind proved harder to measure. The REMS wind sensor was damaged at landing, probably by debris, and the winds derived with a new retrieval point to local circulation as the main driver[7]. During the 2018 global dust storm, the diurnal range of air temperature at Gale fell from about 71 K to 36 K, while the diurnal pressure range rose from about 84 Pa to 124 Pa[8].

The APSS on InSight combined a high-precision pressure sensor, two wind and air-temperature booms of REMS heritage and a magnetometer, mainly to identify atmospheric noise in the records of the seismometer[9]. It sampled pressure at 20 Hz and wind and temperature at 1 Hz almost continuously, and it recorded gravity waves, bores, turbulence, a regional dust storm that amplified the tides and the most intense vortex activity yet observed on Mars. Even on the nearly flat plain of Elysium Planitia, the diurnal cycle of wind direction was controlled by a gentle regional slope[10]. In 400 sols, about 6,000 vortex-induced pressure drops deeper than 0.35 Pa were counted, although no dust devil was imaged[11].

MEDA on Perseverance carries wind, air-temperature, pressure and humidity sensors together with a thermal-infrared sensor and a radiation and dust sensor[12]. Recording in one-hour sessions that alternate between even and odd hours, and thus covering the diurnal cycle every two sols, it observed at Jezero a stable nocturnal inversion, a turbulent daytime regime, abundant vortices and dust devils, and winds driven mainly by local topography[13]. Its pressure tides were weaker than at Gale, a difference that the authors relate to latitude and to the topography and circulation of Gale Crater[14]. The diurnal range of air temperature at 1.45 m was 52–61 K, and the daily

[4]R. M. Haberle et al. "Preliminary interpretation of the REMS pressure data from the first 100 sols of the MSL mission". In: *Journal of Geophysical Research: Planets* 119.3 (2014), pp. 440–453.

[5]S. D. Guzewich et al. "Atmospheric tides in Gale Crater, Mars". In: *Icarus* 268 (2016), pp. 37–49.

[6]M. I. Richardson and C. E. Newman. "On the relationship between surface pressure, terrain elevation, and air temperature. Part I: The large diurnal surface pressure range at Gale Crater, Mars and its origin due to lateral hydrostatic adjustment". In: *Planetary and Space Science* 164 (2018), pp. 132–157.

[7]D. Viúdez-Moreiras et al. "Gale surface wind characterization based on the Mars Science Laboratory REMS dataset. Part I: Wind retrieval and Gale's wind speeds and directions". In: *Icarus* 319 (2019), pp. 909–925.

[8]D. Viúdez-Moreiras et al. "Effects of the MY34/2018 Global Dust Storm as Measured by MSL REMS in Gale Crater". In: *Journal of Geophysical Research: Planets* 124.7 (2019), pp. 1899–1912.

[9]D. Banfield et al. "InSight Auxiliary Payload Sensor Suite (APSS)". In: *Space Science Reviews* 215, 4 (2019).

[10]D. Banfield et al. "The atmosphere of Mars as observed by InSight". In: *Nature Geoscience* 13.3 (2020), pp. 190–198.

[11]A. Spiga et al. "A Study of Daytime Convective Vortices and Turbulence in the Martian Planetary Boundary Layer Based on Half-a-Year of InSight Atmospheric Measurements and Large-Eddy Simulations". In: *Journal of Geophysical Research: Planets* 126.1, e2020JE006511 (2021).

[12]J. A. Rodriguez-Manfredi et al. "The Mars Environmental Dynamics Analyzer, MEDA. A Suite of Environmental Sensors for the Mars 2020 Mission". In: *Space Science Reviews* 217, 48 (2021).

[13]J. A. Rodriguez-Manfredi et al. "The diverse meteorology of Jezero crater over the first 250 sols of Perseverance on Mars". In: *Nature Geoscience* 16 (2023), pp. 19–28.

[14]A. Sánchez-Lavega et al. "Mars 2020 Perseverance Rover Studies of the Martian Atmosphere Over Jezero From Pressure Measurements". In: *Journal of Geophysical Research: Planets* 128.1, e2022JE007480 (2023).

mean was about 6.5 K higher than at Viking Lander 1, which was attributed to the lower albedo of the surface[15].

These studies describe a signal whose components differ in predictability. The diurnal cycle and the tides are forced by insolation and repeat from sol to sol; the seasonal pressure cycle and the dust loading shift the mean state slowly; turbulence and vortices vary within seconds to minutes. The forcing is common to all sites, while the amplitude of the response depends on latitude, topography and surface properties[16]. The literature is, however, diagnostic: it explains the records after the fact, through composite diurnal cycles, harmonic amplitudes and event counts, and does not test how far ahead any component can be predicted. Its comparisons between sites are also sensitive to method, because the stations sample differently and derived statistics depend on detection settings. InSight's vortex rate, for example, was reported as up to five times that of Curiosity in one study[17] and about ten times in another that used a different pressure-drop threshold[18]. No reviewed study compares the three stations on a common time grid or in terms of predictability.

2.3 Physical models and their predictions

Martian weather is predicted with the same tools as terrestrial weather. The GCMs of the Laboratoire de Météorologie Dynamique (LMD) and the University of Oxford share one package of parameterized physics, which covers the radiative effects of carbon dioxide and dust, turbulent mixing in the boundary layer and the condensation of carbon dioxide on the polar caps. The grid-point model was typically run with a spacing of 3.75° in latitude and 5.625° in longitude, a few hundred kilometers at the equator[19]. Output of both models was condensed into the Mars Climate Database (MCD), which the European Space Agency commissioned to support mission design. The database stores seasonal means at twelve times of day for several dust scenarios, together with statistical models of day-to-day variability[20], and a 2024 conference abstract on its version 6.1 lists studies of entry, descent and landing among its current uses[21]. Such climatologies can be accurate: at Jezero, the seasonal evolution of the mean daily pressure predicted by the MCD deviated from the MEDA measurements by at most about 10 Pa[22]. They describe the expected state and its spread for a season and a time of day, however, not the weather of a particular sol.

[15] A. Munguira et al. "Near Surface Atmospheric Temperatures at Jezero From Mars 2020 MEDA Measurements". In: *Journal of Geophysical Research: Planets* 128.3, e2022JE007559 (2023).

[16] Cf. Martínez et al., *op. cit.*; Richardson and Newman, *op. cit.*

[17] Banfield et al., *op. cit.*

[18] Spiga et al., *op. cit.*

[19] F. Forget et al. "Improved general circulation models of the Martian atmosphere from the surface to above 80 km". In: *Journal of Geophysical Research: Planets* 104.E10 (1999), pp. 24155–24175.

[20] S. R. Lewis et al. "A climate database for Mars". In: *Journal of Geophysical Research: Planets* 104.E10 (1999), pp. 24177–24194.

[21] E. Millour et al. "The Mars Climate Database, MCD version 6.1". In: *Europlanet Science Congress 2024*. Conference abstract EPSC2024-516. Copernicus, 2024.

[22] Sánchez-Lavega et al., *op. cit.*

Local circulation requires finer grids. Before Perseverance landed, nine simulations by global and mesoscale models, with grid spacings from about 300 km down to 1.4 km, were compared for Jezero. They agreed on the seasonal pressure cycle, largely because most models had been tuned to the Viking pressure record, and their temperatures differed by up to about 10 K. Their daily pressure cycles differed more, and their winds differed widely, with annual maximum speeds between 11 and 19 $\mathrm{m\,s^{-1}}$. Because of the computational cost, only the coarse simulations covered a full year[23]. At Gale, a mesoscale model matched REMS within the observational uncertainty, but it could not predict the daily mean pressure, which depends on the total mass of the atmosphere, it overestimated the diurnal tide and, in one season, it underestimated daytime winds by a factor of three or more[24].

Weather prediction in the strict sense, a forecast started from current observations, has been studied for Mars at the global scale. The assimilation of orbital temperature and dust retrievals into GCMs produced the Mars Analysis Correction Data Assimilation (MACDA) and Ensemble Mars Atmosphere Reanalysis System (EMARS) reanalyses, which cover parts of Mars Years 24 to 33 on grids of 5° to 6°[25]. Forecasts started from such analyses showed fast, chaotic error growth on time scales of 3 to 10 sols only in northern autumn and winter, and remained coherent for several tens of sols in other seasons. Their skill was measured in the free atmosphere, where orbital data constrain the model best, and the predictability of near-surface conditions was left open[26].

Physical models obey conservation laws, apply where no station exists and can represent conditions never observed at a site. For short-term forecasts at a station, however, the reviewed studies point to two weaknesses. Grids and parameterizations miss part of the local circulation that shapes crater weather, and the predictions are not conditioned on the station's own latest measurements. A data-driven model has the opposite profile: it uses the local record directly but has no physical basis outside the range of its training data. The two approaches are therefore complementary, which is how Chapter 1 frames the forecaster developed in this thesis.

2.4 Data-driven forecasting of Martian weather

Machine learning has been applied to Martian meteorological data mainly for retrieval rather than forecasting, for example to derive winds from the seismic records of InSight[27] or to model

[23] C. E. Newman et al. "Multi-model Meteorological and Aeolian Predictions for Mars 2020 and the Jezero Crater Region". In: *Space Science Reviews* 217, 20 (2021).

[24] J. Pla-Garcia et al. "The meteorology of Gale crater as determined from rover environmental monitoring station observations and numerical modeling. Part I: Comparison of model simulations with observations". In: *Icarus* 280 (2016), pp. 103–113.

[25] L. Montabone et al. "The Mars Analysis Correction Data Assimilation (MACDA) Dataset V1.0". In: *Geoscience Data Journal* 1.2 (2014), pp. 129–139; S. J. Greybush et al. "The Ensemble Mars Atmosphere Reanalysis System (EMARS) Version 1.0". In: *Geoscience Data Journal* 6.2 (2019), pp. 137–150.

[26] P. Rogberg et al. "Assessing atmospheric predictability on Mars using numerical weather prediction and data assimilation". In: *Quarterly Journal of the Royal Meteorological Society* 136.651 (2010), pp. 1614–1635.

[27] A. E. Stott et al. "WindSightNet: The Inter-Annual Variability of Martian Winds Retrieved From InSight's Seismic Data With Machine Learning". In: *Journal of Geophysical Research: Planets* 130.2, e2024JE008695

the relative humidity measured at Gale from the variables of a climate model[28]. Forecasting from station records has received far less attention. The closest precedent is the work of de Cabo García et al. A conference abstract of 2019 described an artificial neural network that predicted the hourly air temperature and pressure of the next sol from those of the current sol. Trained on the first 1,227 of 1,637 REMS sols, it reached a correlation coefficient of 0.990 on the last 410[29]. The journal version of 2022 added the two Viking landers. A small feed-forward network, with the previous sol's local time, air temperature, pressure and solar longitude as inputs, reached a coefficient of determination (R^2) of 0.991 for air temperature on the held-out Curiosity sols, with errors mostly below 0.5 % for temperature and 0.2 % for pressure. Ground temperature, humidity and dust opacity were also examined but, according to the authors, did not improve the network. When the network trained at one Viking lander was applied to the other, the correlation remained high while the errors exceeded 30 %, and the authors concluded that a network cannot be transferred directly between sites[30].

Two further studies forecast in situ quantities. An unreviewed preprint forecast the next sol's atmospheric opacity at the two Mars Exploration Rover sites; its networks transferred well between the rovers, but their accuracy rose sharply only once the previous sol's opacity was included as an input[31]. Priyadarshini and Puri compared five deep networks on the maximum temperatures measured by Curiosity and found a long short-term memory (LSTM) network the most accurate ($R^2 = 0.864$), as measured against the other networks[32].

Since 2025, data-driven forecasting of Martian weather has grown quickly, but almost entirely on the output of reanalyses and models. Uprety et al. compared LSTM, temporal convolutional network (TCN), Transformer, N-BEATS and TiDE models on the OpenMARS reanalysis at the grid box of the InSight landing site. The TCN and TiDE models forecast temperature and pressure one sol ahead most accurately, no model forecast the onset of dust storms, and no reference forecast was reported[33]. Preprints of 2026 describe global emulators of reanalyses: a foundation model pretrained on partial differential equations outperformed persistence on a

(2025).

[28] N. Abdelmoneim et al. "Expanding Mars's Climate Modeling: Interpretable Machine Learning for Modeling Mars Science Laboratory Relative Humidity". In: *The Planetary Science Journal* 5.4, 86 (2024).

[29] A. de Cabo García et al. "MARSchine Learning: Prediction of Mars meteorological variables using artificial neural networks". In: *CPESS 6: VI Meeting of Planetary Science and Solar System Exploration*. Extended abstract. Torrejón de Ardoz: INTA, 2019. URL: https://www.inta.es/export/sites/default/.galleries/cpess6-Descarga/de-Cabo-Garcia-A_MARSchine-Learning-Prediction-of-Mars-meteorological-variables-using-artificial-neural-networks.pdf (accessed 2026-10-07).

[30] A. de Cabo García et al. "Prediction of Mars meteorological variables using artificial neural networks". In: *DYNA New Technologies* 9.1 (2022).

[31] D. Bellutta. *The Fog of War: A Machine Learning Approach to Forecasting Weather on Mars*. 2017. URL: https://arxiv.org/abs/1706.08915 (accessed 2026-10-12).

[32] I. Priyadarshini and V. Puri. "Mars weather data analysis using machine learning techniques". In: *Earth Science Informatics* 14.4 (2021), pp. 1885–1898; for the data set, cf. S. Uprety et al. "Weather Prediction on Mars as a Multivariate Time Series Forecasting Problem". In: *Proceedings of the Workshop on AI-driven Data Engineering and Reusability for Earth and Space Sciences (DARES 2025)*. Vol. 4128. CEUR Workshop Proceedings. 2025, pp. 31–43. URL: https://ceur-ws.org/Vol-4128/paper2.pdf (accessed 2026-10-12).

[33] *Idem*, "Weather Prediction on Mars".

held-out year[34], and the terrestrial GraphCast, applied to Mars without retraining, decayed toward the climatological mean and missed the diurnal cycle until it was fine-tuned[35]. A conference abstract reported preliminary results of a convolutional recurrent network that forecast the column dust opacity one sol ahead with 18 % skill relative to persistence[36], and a roadmap for a Martian foundation model regards rover records as too sparse for training planet-wide models and reserves them for validation[37].

For the purpose of this thesis, these studies share three weaknesses. First, none of the in situ studies whose methods could be checked compared its forecasts with a reference forecast. Because the networks received the previous sol's values at the same local time, a high correlation is expected even from persistence, and correlation measures only potential skill (Section 2.6). Second, the studies by de Cabo García et al. and the opacity preprint forecast one or two variables a single sol ahead, and the development data of the 2022 study were divided at random, with only the last 410 sols held out in time. Third, the more recent work forecasts reanalysis fields on grids of about 300 km, too coarse to resolve the local effects described in Section 2.2[38]. None of the reviewed forecasting studies used MEDA or APSS records.

2.5 Data-driven weather forecasting on Earth

On Earth, machine learning moved within a few years from a benchmark exercise to forecasts that compete with operational numerical weather prediction (NWP). WeatherBench defined a common task on the ERA5 reanalysis, regridded to between 5.625° and about 1.4°, with 500 hPa geopotential and 850 hPa temperature three and five days ahead as headline targets. It listed persistence, climatology and the Integrated Forecasting System (IFS) of the European Centre for Medium-Range Weather Forecasts (ECMWF) as references and stated that a useful system must at least beat persistence and a weekly climatology[39]. Pangu-Weather and GraphCast, both trained on 39 years of ERA5, then outperformed the deterministic IFS forecast at 0.25° resolution: Pangu-Weather on all tested variables when verified against reanalysis[40], and GraphCast on 90 % of 1,380 verification targets, with a ten-day forecast computed in under one minute[41].

This success rests on ERA5, an hourly global reanalysis with a grid spacing of 31 km,

[34] J. Schmude et al. *PDE foundation models are skillful AI weather emulators for the Martian atmosphere*. 2026. URL: https://arxiv.org/abs/2602.15004 (accessed 2026-10-12).

[35] M. L. Carroll et al. *MarsCast: Transfer Learning of AI Weather Foundation Models to Planetary Atmospheres*. 2026. URL: https://arxiv.org/abs/2608.05054 (accessed 2026-10-12).

[36] L. Montabone et al. "Weather on Mars: From Monitoring to Forecasting". In: *Europlanet Science Congress 2026*. Conference abstract EPSC2026-1310. Copernicus, 2026.

[37] S. Roy et al. *Towards a Foundation Model for the Martian Atmosphere*. 2026. URL: https://arxiv.org/abs/2605.28851 (accessed 2026-10-12).

[38] Cf. Richardson and Newman, *op. cit.*; Roy et al., *op. cit.*

[39] S. Rasp et al. "WeatherBench: A Benchmark Data Set for Data-Driven Weather Forecasting". In: *Journal of Advances in Modeling Earth Systems* 12.11, e2020MS002203 (2020).

[40] K. Bi et al. "Accurate medium-range global weather forecasting with 3D neural networks". In: *Nature* 619.7970 (2023), pp. 533–538.

[41] R. Lam et al. "Learning skillful medium-range global weather forecasting". In: *Science* 382.6677 (2023), pp. 1416–1421.

produced by assimilating observations into the IFS and, at the time of its publication, covering the years from 1979 onwards[42]. No comparable data set exists for Mars, whose reanalyses span a few Mars years on grids of 5° to 6° (Section 2.3). What does transfer is the evaluation practice: held-out years, scores for each variable and lead time, and comparison with persistence, climatology and an operational model, which WeatherBench 2 formalized as an open framework with scorecards against the ECMWF forecasts[43].

Station-level forecasting is closer to the Martian setting, and its literature shows the same tension between model complexity and simple references. In very-short-term wind power forecasting, persistence serves as the benchmark and usually beats NWP for the first few hours[44]. At 24 Austrian stations, persistence was more accurate than an ensemble of neural networks fed with observations and NWP output during the first two forecast hours, and the authors concluded that this range needed a different strategy[45]. For air temperature at German stations, a seasonal autoregressive integrated moving average (ARIMA) model and a univariate LSTM network performed well in the first hours but were overtaken by multivariate networks at longer horizons[46]. Other studies omit simple references altogether: a TCN applied to farm weather stations in the United Kingdom was compared only with linear and support-vector regression, over a test period of about three weeks[47].

2.6 Forecasting models and evaluation practice

Gradient boosting builds an additive model stage by stage, each stage fitting a base learner, typically a regression tree, to the negative gradient of the loss of the current model[48]. XGBoost made the method scalable and learns a default direction for missing values at each split[49], and LightGBM made training up to more than twenty times faster with almost the same accuracy[50].

[42] H. Hersbach et al. "The ERA5 global reanalysis". In: *Quarterly Journal of the Royal Meteorological Society* 146.730 (2020), pp. 1999–2049.

[43] S. Rasp et al. "WeatherBench 2: A Benchmark for the Next Generation of Data-Driven Global Weather Models". In: *Journal of Advances in Modeling Earth Systems* 16.6, e2023MS004019 (2024).

[44] A. M. Foley et al. "Current methods and advances in forecasting of wind power generation". In: *Renewable Energy* 37.1 (2012), pp. 1–8.

[45] P. Papazek et al. "Feature selection, ensemble learning, and artificial neural networks for short-range wind speed forecasts". In: *Meteorologische Zeitschrift* 29.4 (2020), pp. 307–322.

[46] D. Kreuzer, M. Munz, and S. Schlüter. "Short-term temperature forecasts using a convolutional neural network – An application to different weather stations in Germany". In: *Machine Learning with Applications* 2, 100007 (2020).

[47] P. Hewage et al. "Temporal convolutional neural (TCN) network for an effective weather forecasting using time-series data from the local weather station". In: *Soft Computing* 24.21 (2020), pp. 16453–16482.

[48] J. H. Friedman. "Greedy function approximation: A gradient boosting machine". In: *The Annals of Statistics* 29.5 (2001), pp. 1189–1232.

[49] T. Chen and C. Guestrin. "XGBoost: A Scalable Tree Boosting System". In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. ACM, 2016, pp. 785–794.

[50] G. Ke et al. "LightGBM: A Highly Efficient Gradient Boosting Decision Tree". In: *Advances in Neural Information Processing Systems*. Vol. 30. Curran Associates, Inc., 2017, pp. 3146–3154. URL: https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html (accessed 2026-10-07).

In the M5 competition on retail sales, practically all of the fifty best teams used LightGBM[51], and a benchmark of 45 tabular data sets, which excluded time series, found tree ensembles more accurate than deep networks on medium-sized data of about 10,000 samples[52].

Deep sequence models learn directly from the history. Recurrent networks with gated memory cells, such as the LSTM[53], were long the default; a review found them competitive with exponential smoothing and ARIMA in many situations, but behind these methods in robustness, efficiency and automation[54]. In an empirical comparison released as a preprint, TCNs, built from causal dilated convolutions with residual connections, outperformed canonical recurrent networks on most of a broad set of sequence-modeling tasks and retained a longer effective memory, although none of the tasks was real-valued forecasting[55]. Larger architectures, such as DeepAR, N-BEATS and the Temporal Fusion Transformer, were developed for and evaluated on collections of many related series, from which they learn shared patterns[56].

Greater complexity has not reliably produced greater accuracy. In the M4 competition, the most accurate methods were combinations, mostly of statistical methods, and a hybrid of exponential smoothing and a recurrent network, whereas none of the pure ML entries beat a simple combination benchmark[57]. In M5, where ML methods won, about 92.5 % of the teams still failed to beat the best simple benchmark[58], and on long-horizon benchmarks a single linear layer outperformed a series of Transformer models[59]. Moreover, the benchmark data, mostly economic, retail, energy and traffic series, differ from Martian records, which are few, long, sampled at high frequency and dominated by one period, the sol. The evidence therefore favors compact and efficient models, compared with simple references, over architectures whose advantages appear on large collections of series.

Evaluation practice has converged on a few rules. Forecasts are evaluated out of sample, on data that follow the training period, and the evaluation is more reliable with rolling forecast

[51] S. Makridakis, E. Spiliotis, and V. Assimakopoulos. "M5 accuracy competition: Results, findings, and conclusions". In: *International Journal of Forecasting* 38.4 (2022), pp. 1346–1364.

[52] L. Grinsztajn, E. Oyallon, and G. Varoquaux. "Why Do Tree-Based Models Still Outperform Deep Learning on Typical Tabular Data?" In: *Advances in Neural Information Processing Systems 35 (Datasets and Benchmarks Track)*. 2022, pp. 507–520.

[53] S. Hochreiter and J. Schmidhuber. "Long Short-Term Memory". In: *Neural Computation* 9.8 (1997), pp. 1735–1780.

[54] H. Hewamalage, C. Bergmeir, and K. Bandara. "Recurrent Neural Networks for Time Series Forecasting: Current status and future directions". In: *International Journal of Forecasting* 37.1 (2021), pp. 388–427.

[55] S. Bai, J. Z. Kolter, and V. Koltun. *An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling*. 2018. URL: https://arxiv.org/abs/1803.01271 (accessed 2026-10-07).

[56] Cf. D. Salinas et al. "DeepAR: Probabilistic forecasting with autoregressive recurrent networks". In: *International Journal of Forecasting* 36.3 (2020), pp. 1181–1191; B. N. Oreshkin et al. "N-BEATS: Neural basis expansion analysis for interpretable time series forecasting". In: *International Conference on Learning Representations (ICLR 2020)*. 2020. URL: https://openreview.net/forum?id=r1ecqn4YwB (accessed 2026-10-07); B. Lim et al. "Temporal Fusion Transformers for interpretable multi-horizon time series forecasting". In: *International Journal of Forecasting* 37.4 (2021), pp. 1748–1764.

[57] S. Makridakis, E. Spiliotis, and V. Assimakopoulos. "The M4 Competition: 100,000 time series and 61 forecasting methods". In: *International Journal of Forecasting* 36.1 (2020), pp. 54–74.

[58] *Idem*, "M5 accuracy competition: Results, findings, and conclusions".

[59] A. Zeng et al. "Are Transformers Effective for Time Series Forecasting?" In: *Proceedings of the AAAI Conference on Artificial Intelligence* 37.9 (2023), pp. 11121–11128.

origins and several test periods[60]. Cross-validation can be valid for stationary series and for purely autoregressive models with uncorrelated errors[61], but Martian records, with their seasonal cycle, dust events and moving rovers, are not stationary. Simple methods such as the naive forecast (persistence) and the seasonal naive forecast serve as benchmarks that a new method must beat to be worth considering[62]. Accuracy relative to such a reference is expressed as a skill score, the fractional reduction of an error measure. In Murphy's formulation the measure is the mean squared error (MSE), and the skill score relative to climatology equals the squared correlation between forecasts and observations only when the forecasts are free of conditional and unconditional bias; a correlation coefficient therefore measures potential rather than actual skill[63]. Murphy later argued that the more accurate of climatology and persistence should serve as the reference, and noted that a linear combination of the two is more accurate than either[64]. Finally, a difference in accuracy between two models requires a test that respects the serial correlation of forecast errors, such as the Diebold–Mariano test[65] or a block bootstrap, which resamples blocks of consecutive observations[66].

2.7 Transfer of models between sites

A model applied at a site other than its training site faces dataset shift: the joint distribution of inputs and targets differs between the training and the test data[67]. Transfer learning addresses such shifts when the source and target domains are related; when they are not, transfer can degrade performance, an effect known as negative transfer[68]. A model applied to a new site without any data from that site is, in these terms, evaluated out of domain.

Two lines of work on Earth indicate what can be expected. Montero-Manso and Hyndman showed that a single global model fitted to a group of series can produce the same forecasts as separate local models without assuming that the series are similar, and they found global models more accurate on standard benchmark collections. Their benchmarks, however, excluded

[60] L. J. Tashman. "Out-of-sample tests of forecasting accuracy: an analysis and review". In: *International Journal of Forecasting* 16.4 (2000), pp. 437–450.

[61] C. Bergmeir and J. M. Benítez. "On the use of cross-validation for time series predictor evaluation". In: *Information Sciences* 191 (2012), pp. 192–213; C. Bergmeir, R. J. Hyndman, and B. Koo. "A note on the validity of cross-validation for evaluating autoregressive time series prediction". In: *Computational Statistics & Data Analysis* 120 (2018), pp. 70–83.

[62] Cf. R. J. Hyndman and G. Athanasopoulos. *Forecasting: Principles and Practice*. 3rd ed. Melbourne: OTexts, 2021. URL: `https://otexts.com/fpp3/` (accessed 2026-10-07).

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[67] Cf. J. Quiñonero-Candela et al., eds. *Dataset Shift in Machine Learning*. Neural Information Processing Series. Cambridge, MA: MIT Press, 2009.

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weather data and forecast only series seen in training[69]. In hydrology, LSTM networks trained on subsets of 531 catchments and applied to the catchments withheld from training reached a median Nash–Sutcliffe efficiency of 0.69, higher than a conceptual model calibrated in each catchment (0.64) but lower than the same networks trained with all catchments (0.74)[70]. This transfer relied on hundreds of training catchments and on static descriptors of each one. Station forecasting points the other way: the relations between variables differed so much between Austrian stations that each station received its own model, and adding data from similar neighboring stations brought only a small gain[71].

In the reviewed literature, transfer between Martian sites has been tested only for two pairs of sites, with simple networks forecasting one sol ahead. Networks trained at one Viking lander lost much of their accuracy at the other[72], whereas opacity forecasts transferred well between the two Mars Exploration Rovers[73]. With three long station records and no comparable set of site descriptors, neither the many-series argument nor descriptor-based transfer applies to Mars directly. The transfer error is therefore less a test of a deployable model than a measure of how much of the learned structure is specific to one site, which is the question that RQ3 asks.

2.8 Synthesis, research gap and position of the thesis

Table 2.1 compares the main forecasting approaches reviewed in this chapter. Three patterns stand out. First, the physical approaches describe the expected state or predict global fields, but none is conditioned on the latest measurements of a station. Second, the data-driven forecasts of Martian weather use either station records without a reference forecast or reanalysis fields that do not resolve the local circulation. Third, the terrestrial studies that compare with simple references show that persistence is hard to beat at the shortest horizons and that successful transfer to new sites relied on many training sites.

Three gaps follow from the reviewed literature, one for each research question of Chapter 1:

G1 No data-driven forecast of Martian station records has been evaluated against reference forecasts, with skill scores and confidence intervals, across several variables and horizons, and no forecasting study has used the records of MEDA or APSS (RQ1).

G2 The value of secondary measurements as forecast inputs has been examined only in passing, in the report that ground temperature, humidity and opacity did not improve a next-sol network, although the observational studies show that the dust loading changes both the diurnal temperature range and the tides (RQ2).

G3 Transfer between Martian sites has been tested only for two pairs of sites, one sol ahead

[69] P. Montero-Manso and R. J. Hyndman. "Principles and algorithms for forecasting groups of time series: Locality and globality". In: *International Journal of Forecasting* 37.4 (2021), pp. 1632–1653.

[70] F. Kratzert et al. "Toward Improved Predictions in Ungauged Basins: Exploiting the Power of Machine Learning". In: *Water Resources Research* 55.12 (2019), pp. 11344–11354.

[71] Papazek et al., *op. cit.*

[72] de Cabo García et al., *op. cit.*

[73] Bellutta, *op. cit.*

and without reference forecasts. Transfer among Gale, Elysium Planitia and Jezero at hourly horizons, including a model pooled across sites, has not been measured (RQ3).

Table 2.1: Forecasting approaches relevant to Martian surface weather

Approach	Data	Forecast target	Reference and metric	Main limitation for station forecasting
Mars Climate Database[74]	Multiannual GCM runs, 3.75° grid	Mean state and variability by season and time of day	Comparison with lander records	Not conditioned on current observations
Multi-model prediction for Jezero[75]	Nine simulations, 1.4–300 km grids	Seasonal and daily cycles, before landing	Comparison between models	Wide disagreement on winds; high computational cost
Next-sol neural network[76]	REMS hourly and Viking daily means	Air temperature and pressure, one sol ahead	Correlation, MSE, R^2; no reference forecast	Random split of development data; one horizon
Deep networks on a Mars reanalysis[77]	OpenMARS at the InSight grid box	Surface variables, one sol ahead	MAE, RMSE; no reference forecast	Forecasts a reanalysis on a 5° grid, not a station
Global ML weather model[78]	ERA5, 39 years, 0.25° grid	Global fields, up to ten days	Operational IFS, 1,380 targets	Needs decades of reanalysis
Station-level neural networks[79]	Austrian stations and NWP output	Wind speed, 1–40 h	Persistence, NWP, statistical post-processing	Persistence better in the first two hours
LSTM for ungauged catchments[80]	531 catchments with static descriptors	Daily streamflow at unseen sites	Conceptual model calibrated at each site	Needs many sites and site descriptors
This thesis (Chapter 6)	MEDA, REMS and APSS station records	Air temperature, pressure and wind speed, 1–48 h	Persistence and time-of-sol climatology; skill; bootstrap intervals	Three sites, no site descriptors

Source: own elaboration based on the cited works.

The thesis addresses these gaps without proposing a new architecture. It uses two established models that the reviewed literature supports for this setting: gradient-boosted trees, which are fast, handle missing values and perform well on tabular data of moderate size, and a TCN, an efficient

[74] Lewis et al., *op. cit.*
[75] Newman et al., *op. cit.*
[76] de Cabo García et al., "MARSchine Learning"; *idem*, "Prediction of Mars meteorological variables".
[77] Uprety et al., *op. cit.*
[78] Lam et al., *op. cit.*
[79] Papazek et al., *op. cit.*
[80] Kratzert et al., *op. cit.*

sequence model with a long effective memory. Larger architectures and global weather models are not used, because they were designed for large collections of series or for gridded global states, a setting that three station records do not provide, and because the study was designed for a consumer laptop GPU (Chapter 1). The models are evaluated with the practice of terrestrial forecasting: chronological splits, persistence and a time-of-sol climatology as references, skill scores and, for the paired and cross-site comparisons, confidence intervals from a block bootstrap over sols. What the thesis takes from existing work is therefore its models and its evaluation methods. Its own contribution is their application to the station records of Jezero, Gale and Elysium Planitia, the paired test of secondary signals and the measured transfer between the three sites. The hypotheses and success criteria that follow from these gaps are stated in Chapter 6.

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